

HW 2.

Due: Oct. 12, 2026 at 3:30pm.

Please use python as your programming language.

Problem 1. Prove that $I - 2P$ is orthogonal if P is an orthogonal projection matrix. Give a geometric interpretation.

Problem 2. Consider the matrices

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Answer the following questions by hand calculation.

- a) What is the orthogonal projection matrix P onto $\text{range}(A)$, and what is the image under P of the vector $(1, 2, 3)^T$? Using any method you like, determine (on paper) a reduced QR factorization $A = \hat{Q}\hat{R}$ and a full QR factorization $A = QR$.
- b) Repeat a) for B .

Problem 3. Let x, y, u , and v be nonzero vectors in $\mathbb{R}^3$ with the property that x and y are linearly independent and so are u and v . Consider the two planes in $\mathbb{R}^3$

$$P_1 = \langle x, y \rangle, \quad P_2 = \langle u, v \rangle$$

where $\langle x_1, x_2, \dots, x_n \rangle$ is the subspace spanned by $x_1, x_2, \dots, x_n$. Suppose we wish to find a nonzero vector $v \in \mathbb{R}^3$ that lies in the intersection $P = P_1 \cap P_2$. Devise a method for solving this problem by reducing it to the computation of QR factorizations of three 3×2 matrices.

Problem 4.

- a) Write python functions named “mgs(A)” and “cgs(A)” that computes a reduced QR factorization $A = \hat{Q}\hat{R}$ of $A \in \mathbb{R}^{m \times n}$ with $m \geq n$ using classical and modified Gram-Schmidt orthogonalization, respectively. The output variables are a matrix $Q \in \mathbb{R}^{m \times n}$ with orthonormal columns and a triangular matrix $R \in \mathbb{R}^{m \times n}$.
- b) Consider

$$A = \begin{pmatrix} 1 & 1 & 1 \\ \varepsilon & \varepsilon & 0 \\ \varepsilon & 0 & \varepsilon \end{pmatrix}$$

with $\varepsilon = 10^{-8}$. Apply your functions (msg and cgs) to A to find orthonormal column vectors. Check whether or not the resulting column vectors are orthonormal by computing $Q^T Q$.

- c) Consider a matrix B

$$B = \varepsilon \text{np.eye}(n) + \text{scipy.linalg.hilbert}(n)$$

with $n = 200$ and $\varepsilon = 0.0001$. Apply your functions (msg and cgs) to B to find orthonormal column vectors. Check whether or not the resulting column vectors are orthonormal by computing $\|I - Q^T Q\|_2$.

Problem 5. Consider A , the $m \times m$ matrix (a Toeplitz matrix) with 1 on the main diagonal, 2 on the first superdiagonal, and 0 everywhere else.

- a) What are the eigenvalues, determinant, and rank of A ?
- b) What is A^{-1} ?
- c) Give a nontrivial upper bound on σ_m , the m th singular value of A .